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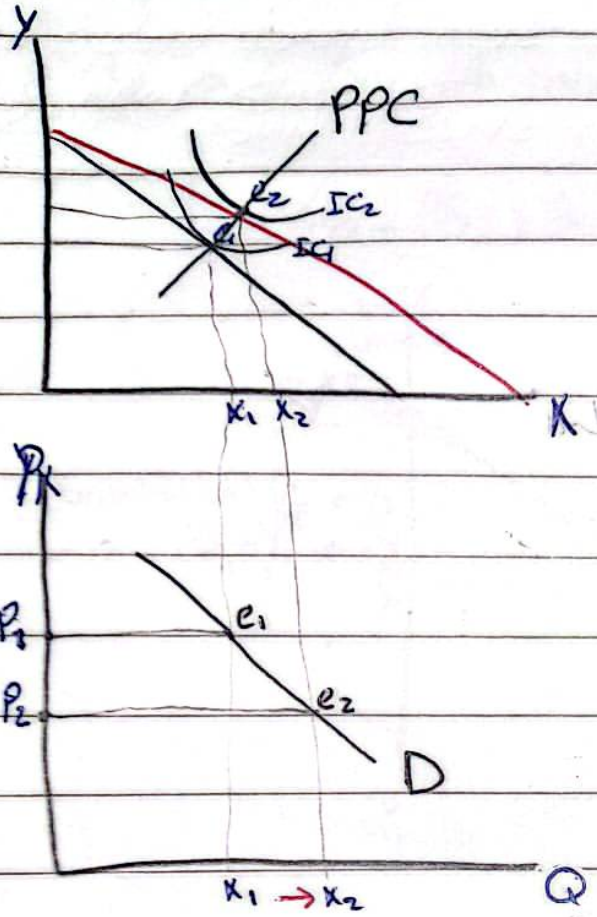
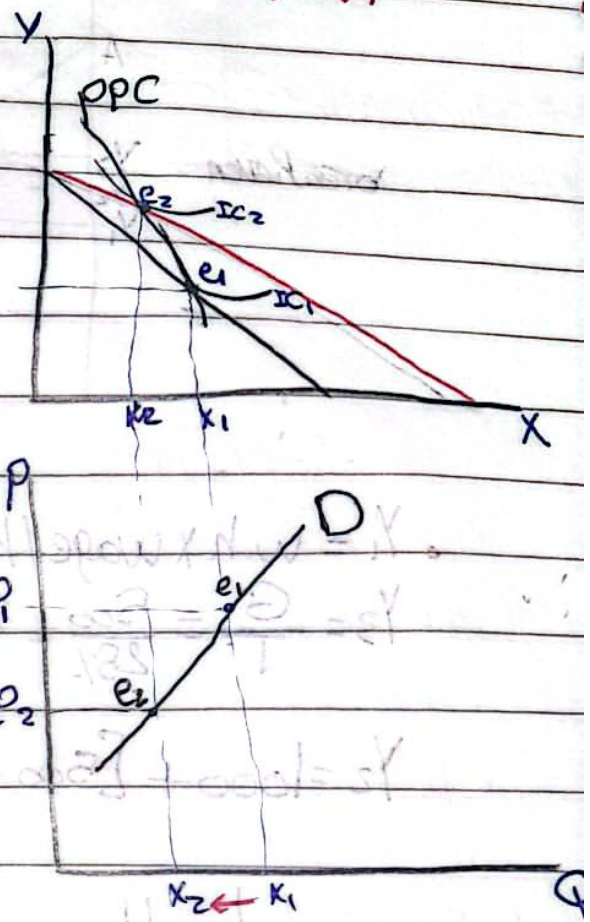
deriving demand curve

incase of Normal Good
($P_X \downarrow$)



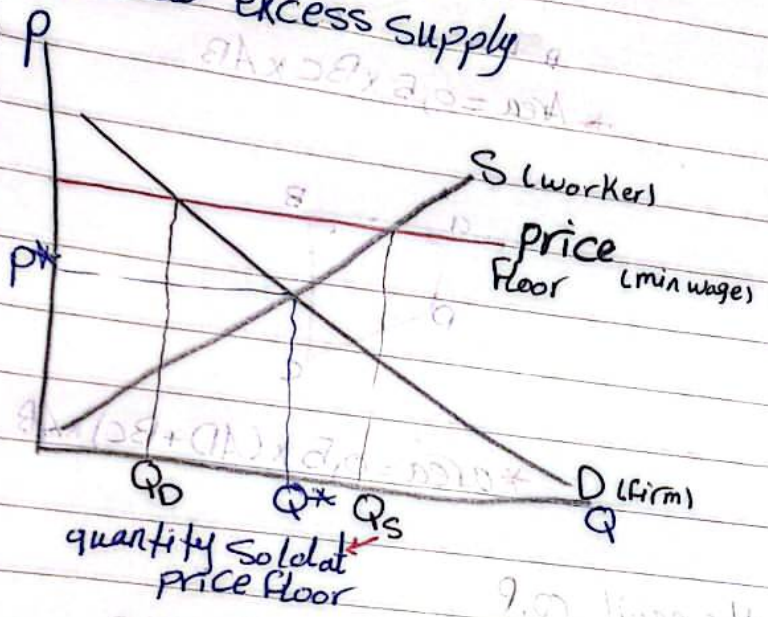
incase of giffen good
($P_X \downarrow$)



Gov. intervention

price floor

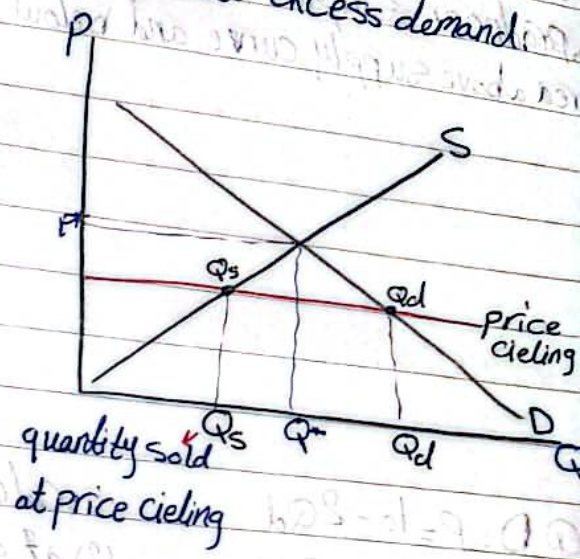
* minimum price set by Gov.
* above equil price.
* Create excess supply



* $Q_S > Q_d$

price cieling

* maximum price set by gov.
* below equil price.
* Create excess demand



* $Q_d > Q_s$

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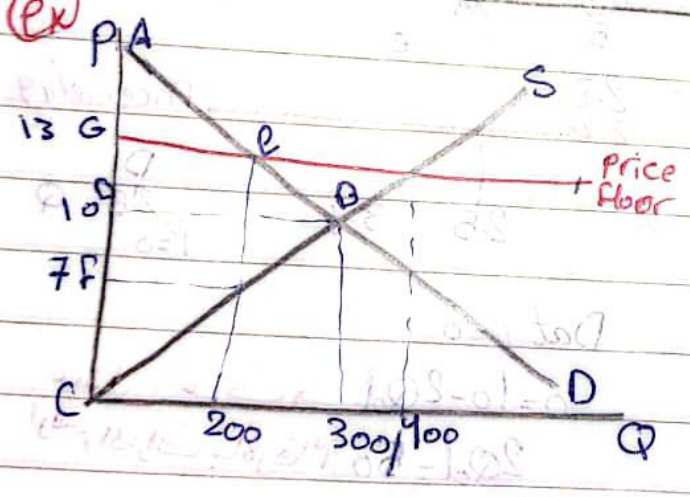
ex



- befor gov intervention ($P=10$)
  - $CS = \frac{1}{2} \times 300 \times (20-10) = 1500$ (area ABD)
  - PS $\frac{1}{2} \times 300 \times (10-0) = 1500$ (area DBC)
  - $TS = 1500 + 1500 = 3000$
- after gov intervention ($P=13$)
  - $CS = \frac{1}{2} \times 200 \times (20-13) = 700$ (area AeG)
  - $PS = \frac{1}{2} \times [(13-7+13-0)] \times 200 =$ (area GeFC)
    $= \frac{1}{2} \times 19 \times 200 = 1900$
  - $TS = 700 + 1900 = 2600$

* Dead Weight Loss =
TS before − TS after =
3000 − 2600 = 400 (area eBF)

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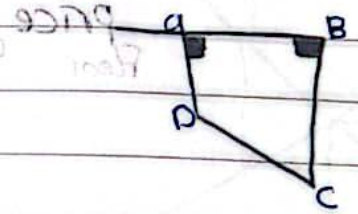
*Note:

→ Consumer surplus:
area below demand curve and above price

→ Producer surplus
area above supply curve and below price

Note

* Area = 0,5 × BC × AB



* area = 0,5 × (AD + BC) × AB

Q) D: $P = 10 - 2Q_d$
S: $P = 1 + 0,5Q_s$

(1) Calculate the equil Q, P
(2) if gov put price cieling = 2.25 what is the new Q
(3) Calculate CS, PS at initial equil price
(4) PS, CS after price cieling

(1) $D = S$

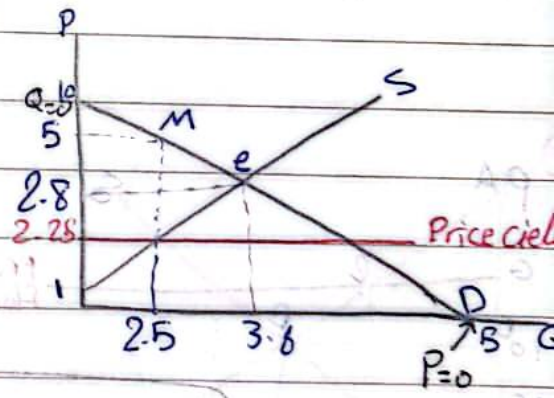
$10 - 2Q = 1 + 0,5Q$

$10 - 1 = 2.5Q$

$9 = 2.5Q$

$\boxed{Q^* = \frac{9}{2.5} = 3.6}$

$\boxed{D: P^* = 10 - 2 * 3.6 = 2.8}$



(2) at $P = 2.25$

S: $2.25 = 1 + 0,5Q_s$

$1,25 = 0,5Q_s$

$\boxed{Q_s = 2.5}$

S at $Q = 0$

$P = 1 + 0,5 \times 0$

$P = 1$

D at $P = 0$

$0 = 10 - 2Q_d$

$2Q_d = 10$

$Q_d = 5$

D at $Q = 0$

$P = 10 - 2 \times 0$

$P = 10$

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(3) at $P = 2.8$

$CS = 0,5 \times 3.6 \times (10 - 2.8) = 12.96$

$PS = 0,5 \times 3.6 \times (2.8 - 1) = 3.24$

$TS = 12.96 + 3.24 = 16,2$

(4) $CS = 0,5 \times [(10 - 2.25) + (5 - 2.25)] \times 2.5$

$= 0,5 \times 10.5 \times 2.5$

$= 13,125$

$PS = 0,5 \times 2.5 \times (2.25 - 1)$

$= 1.56$

$TS = 13.125 + 1.56 = 14.68$

ded weigt loss $= 16,2 - 14.68 = 1.52$

price at M

عشان النقطة عليها

$Q = 2.5 \rightarrow$ sub in D equation

$P = 10 - 2Qd$

$P = 10 - 2 \times 2.5 = 5$